

- 65) State whether the function  $f(x) = \frac{3^x + 3^{-x}}{2}$  is odd or even.
- 66) If  $f(x) = 4x^4 + 2x^2 + 1 - 3\cos^2 x - 5\sin^2 x$ , show that  $f(x)$  is an even function.
- 67) If  $f(x) = 4x^4 + 3\cos x + x \cdot \sin x + 1$  show that the function is even.
- 68) If  $f(x) = \log\left(x + \sqrt{1+x^2}\right)$ , show that the function is odd.
- 69) If  $f(x) = \frac{\cos x}{1 + \sin^2 x}$ , determine whether it is an odd or an even function.
- 70) Show that  $f(x) = x^3 + 3\sin x + x$  is an odd function.
- 71) If  $f(x) = \frac{3x-4}{1+x^2}$ , show that  $f(x)$  is an odd function.

**Ex. 9.2.1 :** If  $f(x) = \frac{x^3 + 1}{x^2 + 1}$ , then find  $f(-3)$ ,  $f(-1)$

**Ex. 9.2.2 :** If  $f(x) = 16^x + \log_2 x$  then find  $f\left(\frac{1}{4}\right)$   
and  $f\left(\frac{1}{2}\right)$

Also find the value of  $f\left(\frac{1}{4}\right)^2$ .

**S-13, S-16, 4 Marks, W-13, W-15, W-16, 2 Marks**

**Ex. 9.2.3 :** If  $f(x) = x^3 + x$ , find  $f(1) + f(2)$

**S-11, 2 Marks**

**Ex. 9.2.4 :** If  $f(x) = \frac{x^2 - 3x + 1}{x - 1}$ , find  $f(-2) + f\left(\frac{1}{3}\right)$

**Ex. 9.2.5 :** If  $f(x) = \frac{x^2 + 9}{\sqrt{x - 3}}$ , find  $f(4) + f(5)$

**S-12, 2 Marks**

**Ex. 9.2.6 :** If  $f(x) = ax^2 + bx + 3$  and  $f(1) = 4$ ,  $f(2) = 11$ ,

Find  $a$  and  $b$ .

**S-14, 4 Marks**

**Ex. 9.2.7 :** If  $f(x) = x^3 + 1$  and  $t = y + 2$  then find  $f(t)$ .

**Ex. 9.2.8 :** If  $y = f(x) = \frac{x + 1}{x - 1}$ ,  $x \neq 1$  then show that,

$x = f(y)$ .

**S-2010, 4 Marks**

**Ex. 9.2.9 :** If  $y = f(x) = \frac{x - 5}{5x - 1}$  then show that  $f(y) = x$

**W-2007, 4 Marks**

**Ex. 9.2.10 :** If  $f(x) = \frac{x + 5}{3x - 4}$  and  $t = \frac{5 + 4x}{3x - 1}$  show that

$f(t) = x$ .

**W-2008, 2 Marks**

**Ex. 9.2.11 :** If  $f(x) = x^2 + 3$  then find the value of  $x$  for which  $f(x) = f(2x + 1)$ .

**S-16, 4 Marks**

2] If  $f(x) = \log(x + \sqrt{1+x^2})$ , show that the function is odd.

→ given:  $f(x) = \log(x + \sqrt{1+x^2})$

replacing the argument  $x$  by  $-x$

$$\therefore f(-x) = \log(-x + \sqrt{1+(-x)^2})$$

$$= \log(-x + \sqrt{1+x^2})$$

$$\therefore f(-x) = \log(\sqrt{1+x^2} - x)$$

Multiplying and dividing by  $(\sqrt{1+x^2} + x)$

$$\therefore f(-x) = \log \left\{ (\sqrt{1+x^2} - x) \times \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2} + x} \right\}$$

$$= \log \left\{ \frac{1+x^2 - x^2}{\sqrt{1+x^2} + x} \right\} \dots (a-b)(a+b) = a^2 - b^2$$

⑥

$$= \log \left( \frac{1}{\sqrt{1+x^2} + x} \right)$$

$$= -\log (\sqrt{1+x^2} + x) \quad \dots \because \log \frac{1}{a} = -\log_e a$$

$$= -f(x)$$

$$\therefore f(-x) = -f(x)$$

$\therefore$  It is an odd function .

⑦

3] If  $f(x) = 4x^4 + 3\cos x + x \cdot \sin x + 1$ , show that the function is